Linear functions — Part 2 answer key

#	ID	Answer	Difficulty
2.1	17c37dcf-800e-41d5-8367-1ccfbfa33af2	A	Medium
2.2	f8b5e794-72b0-4393-8d1c-cd433c069575	3	Medium
2.3	d9f57fdb-9af6-41d0-9c42-2de71bdeab1f	B	Medium
2.4	70799514-1e68-4ec1-9002-6b0e1165cbb0	2	Medium
2.5	83a73190-62d2-4465-82e0-b6794ffbddf6	C	Medium
2.6	5c630c81-9ef1-4180-821f-1dd4851610cf	D	Medium
2.7	028813-DC	?	Medium
2.8	d243de3a-1370-41b8-a350-5396b80263a0	A	Medium
2.9	47f92079-253b-46b3-a924-90dc197bbd5c	B	Medium
2.10	89816479-b500-4fe9-b80f-8f9fea007b9f	C	Medium
2.11	f44d3920-a7ed-4c31-815c-11dbfe887d43	C	Medium
2.12	027788-DC	?	Medium
2.13	607e02b9-ac5b-48d2-8d14-a74f33743cd9	D	Medium
2.14	718a67c3-595c-4af9-b1c7-38d2317665f0	C	Medium
2.15	08375-DC	A	Medium
2.16	6033a1e4-007d-42e9-ada3-25ad35a47e81	7	Medium
2.17	32ff6707-691a-4b5c-a8ba-41a26d2c8ad0	D	Medium
2.18	ceae23d4-1965-4201-ac8f-3dbe916cc8e3	D	Medium
2.19	10521-DC	A	Medium
2.20	7fedd1ee-8dd3-449f-8f4a-8154d1d099ed	C	Medium
2.21	06a91641-79ae-4887-9807-53c2c9859f93	D	Medium
2.22	a76b700a-7987-4650-b3bd-3b739ce06f18	A	Medium

2.23	d6701d8a-79c5-4cbd-b549-dcd4b570f33b	D	Medium
2.24	08691-DC	C	Medium
2.25	022313-DC	B	Medium
2.26	77e2c729-a66a-47de-9258-130399ad202e	A	Medium
2.27	14854-DC	A	Medium
2.28	070898-DC	B	Medium
2.29	75be103e-3a39-4793-99d5-2ecb57774a84	C	Medium
2.30	0478e0b5-4347-4ca3-81d3-9d8f3d89e005	A	Medium
2.31	a60d5936-6770-4a18-b3b7-b21e68aadf5a	C	Medium
2.32	022bc7aa-cc1a-41cc-8856-f626e8d243ac	A	Medium
2.33	940b3aad-7c94-4109-8fe4-ef6351e23009	C	Medium
2.34	0beb2470-2acd-4279-917c-0e0172947d97	9	Medium
2.35	3ad33584-7909-480c-b2c2-e9092550290d	C	Medium
2.36	513fd8bd-86d1-4102-aea1-0a8bc65de5dd	B	Medium
2.37	6bf54304-ff33-4fc3-8ab3-b9c8c115f618	C	Medium
2.38	ee5be81f-8eae-4f80-8e45-e6642db5143d	D	Medium
2.39	ef6e2e6b-9410-44c5-86c6-4cce2bdace80	241	Medium
2.40	acd27116-7079-469e-b599-5c238ac3e563	39	Medium
2.41	ce40ca76-4f75-4034-9e39-b6c214f31eea	B	Medium
2.42	025468-DC	C	Medium
2.43	9a75850f-42ca-4792-94aa-fe7d5258a84c	2850	Medium
2.44	05013-DC	B	Medium
2.45	9a64af20-3d37-416c-9771-0ce29547dd42	-42	Medium
2.46	029018-DC	D	Medium
2.47	0d590b1e-c165-452b-bd68-2c7ae4a0a55d	67	Medium
2.48	070611-DC	?	Medium

2.49	f0bbaf49-87a8-4fb3-b54f-eadea2dca1a9	D	Medium
2.50	74ab7488-2efa-4e42-86ec-970c72177393	774	Medium
2.51	b64bd9d3-56b1-45f6-b301-f29dfd4eb579	11	Medium
2.52	461f9ba1-a36e-4795-9b6e-093390fd8977	A	Medium
2.53	11b06370-1f90-4553-8b54-c1ebec99dcad	A	Medium
2.54	f268654f-d968-4474-abc8-911195b0c902	.0465, 2/43	Hard
2.55	e47cdf24-1ef9-4aa4-bc46-70b25da6e45e	515	Hard
2.56	5bba5291-1a05-43fc-bef6-75ab59bb1c60	B	Hard
2.57	025812-DC	A	Hard
2.58	1a8089dd-7f49-40bb-9cf4-9d9040daa84f	A	Hard
2.59	021098-DC	B	Hard
2.60	f3a8f6e5-3640-4af9-a194-e13afbea670e	D	Hard
2.61	4c14e865-e027-430a-9613-c80acd0346bf	3331	Hard
2.62	6ea75b24-df55-4b56-a799-54ee041ec141	A	Hard
2.63	9757792c-41ae-4f55-afe7-6d7d77147ac6	B	Hard
2.64	c0070404-51d0-404d-88ce-75aa7e687a20	B	Hard
2.65	2d19a50f-046d-44a2-92f8-a10cf81a3e3c	A	Hard
2.66	dfe2986a-18ec-40e9-8bd5-2f33f9975a35	500	Hard
2.67	1f9e0de2-d5a7-4f67-9e59-62315da247f3	257	Hard
2.68	bb4ec6e5-ecea-4bc4-a6d2-40a50bb5e60f	A	Hard
2.69	09384-DC	C	Hard
2.70	029517-DC	B	Hard
2.71	ff095f0f-b8e2-470c-903c-78999ec17213	A	Hard
2.72	700cd854-082d-4c3c-a2da-017482d48062	A	Hard
2.73	704bbab2-6312-41d6-a1ca-affd16b16c5a	C	Hard
2.74	09335-DC	D	Hard

2.75	ea18109c-11ea-4e1e-a907-b4a77b4a93e1	A	Hard
2.76	839b7fb0-37a1-46d9-9dce-f3af7e85c97e	-12	Hard
2.77	4959c26d-16dc-4df2-8bb7-92bd72d87aa5	A	Hard
2.78	d70b4faa-5ce1-4664-ae31-432dc125273a	D	Hard
2.79	5904d2e1-1ae5-40cf-915f-050b6b6d3111	D	Hard
2.80	b56fbc38-168f-4197-8169-fd6b32162d42	D	Hard
2.81	08486-DC	D	Hard
2.82	b09c918e-97c2-4a8b-9406-9ed217790fb4	16.25, 65/4	Hard
2.83	4e317fbb-cdce-4c3a-acc2-5f1f342d3af0	D	Hard
2.84	b054f030-a6ac-4c84-8808-fddfd6d758ba	A	Hard
2.85	c16f36d3-645b-4e76-856d-2cc2c923b53c	B	Hard

Rationales

▼ 2.1 — A

Choice A is correct. It's given that the pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch psi. It's also given that for each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. The total pressure, in psi, exerted on the scuba diver at x feet below sea level can be represented by the expression $0.44x + 14.70$. Substituting 105 for x in this expression yields $0.44(105) + 14.70$, or 60.90. Therefore, the total pressure exerted on the scuba diver at 105 feet below sea level is 60.90 psi.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pressure, in psi, exerted on the scuba diver at sea level, not at 105 feet below sea level.

Choice D is incorrect. This is the rate by which the pressure, in psi, exerted on the scuba diver increases for each foot the scuba diver descends below sea level.

▼ 2.2 — 3

The correct answer is 3. Substituting 5 for fx in the given function yields $5 = \frac{9}{7}x + \frac{8}{7}$. Multiplying each side of this equation by 7 yields $75 = 7\frac{9}{7}x + \frac{8}{7}$, or $35 = 9x + 8$. Subtracting 8 from each side of this equation yields $27 = 9x$. Dividing each side of this equation by 9 yields $3 = x$. Therefore, $fx = 5$ when the value of x is 3.

▼ 2.3 — B

Choice B is correct. It's given that the function $f(x) = 55.20 - 0.16x$ gives the estimated surface water temperature, in degrees Celsius, of a body of water on the xth day of the year. Substituting 326 for x in the given function yields $f(326) = 55.20 - 0.16(326)$, which is equivalent to $f(326) = 55.20 - 52.16$, or $f(326) = 3.04$. Therefore, the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year is 3.04.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the rate of change, in degrees Celsius per day, of the estimated surface water temperature.

Choice D is incorrect. This is the change, in degrees Celsius, in the estimated surface water temperature over 326 days.

▼ 2.4 — 2

The correct answer is 2. It's given that function f is defined by $fx = 2x + 3$. Therefore, the equation representing the graph of $y = fx$ in the xy-plane is $y = 2x + 3$, and the graph is a line. For a linear equation in the form $y = mx + b$, m represents the slope of the line. Since the value of m in the equation $y = 2x + 3$ is 2, the slope of the line defined by function f is 2. It's given that line j is parallel to the line defined by function f. The slopes of parallel lines are equal. Therefore, the slope of line j is also 2.

▼ 2.5 — C

Choice C is correct. An equation that defines a linear function f can be written in the form $fx = mx + b$, where m represents the slope and b represents the y-intercept, $0, b$, of the line of $y = fx$ in the xy-plane. The function $fx = -5x + 30$ is linear. Therefore, the graph of the given function $y = fx$ in the xy-plane has a y-intercept of $0, 30$. It's given that fx gives the approximate volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. It follows that the y-intercept of $0, 30$ means that Caleb had approximately 30 fluid ounces of juice remaining after making 0 popsicles. In other words, Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

Choice A is incorrect. This is an interpretation of the slope, rather than the y-intercept, of the graph of $y = fx$ in the xy-plane.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.6 — D

Choice D is correct. An equation defining h can be written in the form $y = mx + b$, where $y = hx$, m represents the slope of the graph of $y = hx$ in the xy-plane, and b represents the y-coordinate of the y-intercept of the graph. It's given that $h28 = 15$ and $h26 = 22$. It follows that the points $28, 15$ and $26, 22$ are on the graph of $y = hx$ in the xy-plane. The slope can be found by using any two points, x_1, y_1 and x_2, y_2, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $28, 15$ and $26, 22$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{22 - 15}{26 - 28}$, or $m = -\frac{7}{2}$. Substituting $-\frac{7}{2}$ for m and $28, 15$ for x, y in the equation $y = mx + b$ yields $15 = -\frac{7}{2}28 + b$, or $15 = -98 + b$. Adding 98 to both sides of this equation yields $113 = b$. Substituting $-\frac{7}{2}$ for m and 113 for b in the equation $y = mx + b$ yields $y = -\frac{7}{2}x + 113$. Since $y = hx$, it follows that the equation that defines h is $hx = -\frac{7}{2}x + 113$.

Choice A is incorrect. For this function, $h26 = \frac{109}{7}$, not $h26 = 22$.

Choice B is incorrect. For this function, $h28 = 105$, not $h28 = 15$, and $h26 = \frac{739}{7}$, not $h26 = 22$.

Choice C is incorrect. For this function, $h28 = -75$, not $h28 = 15$, and $h26 = -68$, not $h26 = 22$.

▼ 2.7 — ?

The correct answer is 3500. The given information includes a cost, $800, to produce 100 shirts. Substituting $c(x) = 800$ and $x = 100$ into the given equation yields $800 = m \cdot 100 + 500$. Subtracting 500 from both sides of the equation yields $300 = m \cdot 100$. Dividing both sides of this equation by 100 yields $3 = m$. Substituting the value of m into the given equation yields $c(x) = 3x + 500$. Substituting 1000 for x in this equation and solving for $c(x)$ gives the cost of 1000 shirts: $3(1000) + 500$, or 3500.

▼ 2.8 — A

Choice A is correct. The given graph is a line, and the slope of a line is defined as the change in the value of y for each increase in the value of x by 1. It's given that y represents

the total cost, in dollars, and that x represents the number of games. Therefore, the change in the value of y for each increase in the value of x by 1 represents the change in total cost, in dollars, for each increase in the number of games by 1. In other words, the slope represents the cost, in dollars, per game. The graph shows that when the value of x increases from 0 to 1, the value of y increases from 100 to 125. It follows that the slope is 25, or the cost per game is $\$25$. Thus, the best interpretation of the slope of the graph is that each game costs $\$25$.

Choice B is incorrect. This is an interpretation of the y-intercept of the graph rather than the slope of the graph.

Choice C is incorrect. The slope of the graph is the cost per game, not the cost of the video game system.

Choice D is incorrect. Each game costs $\$25$, not $\$100$.

▼ 2.9 — B

Choice B is correct. It's given that the graph of the linear function h, where $y = hx$, passes through the points $7, 21$ and $9, 25$ in the xy-plane. An equation defining h can be written in the form $y = mx + b$, where $y = hx$, m represents the slope of the graph in the xy-plane, and b represents the y-coordinate of the y-intercept of the graph. The slope can be found using any two points, x_1, y_1 and x_2, y_2, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $7, 21$ and $9, 25$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{25 - 21}{9 - 7}$, which is equivalent to $m = \frac{4}{2}$, or $m = 2$. Substituting 2 for m and $7, 21$ for x, y in the equation $y = mx + b$ yields $21 = 27 + b$, or $21 = 14 + b$. Subtracting 14 from each side of this equation yields $7 = b$. Substituting 2 for m and 7 for b in the equation $y = mx + b$ yields $y = 2x + 7$. Since $y = hx$, it follows that the equation that defines h is $hx = 2x + 7$.

Choice A is incorrect. For this function, the graph of $y = hx$ in the xy-plane would pass through $7, 0$, not $7, 21$, and $9, 1$, not $9, 25$.

Choice C is incorrect. For this function, the graph of $y = hx$ in the xy-plane would pass through $7, 70$, not $7, 21$, and $9, 84$, not $9, 25$.

Choice D is incorrect. For this function, the graph of $y = hx$ in the xy-plane would pass through $7, 88$, not $7, 21$, and $9, 106$, not $9, 25$.

▼ 2.10 — C

Choice C is correct. It's given that there is a linear relationship between a plumber's hours of labor, h, and the plumber's total charge fh, in dollars. It follows that the relationship can be represented by an equation of the form $fh = mh + b$, where m is the rate of change of the function f and b is a constant. The rate of change of f can be calculated by dividing the difference in two values of fh by the difference in the corresponding values of h. Based on the values given in the table, the rate of change of f is $\frac{285 - 155}{3 - 1}$, or 65. Substituting 65 for m in the equation $fh = mh + b$ yields $fh = 65h + b$. The value of b can be found by substituting any value of h and its corresponding value of fh for h and fh, respectively, in this equation. Substituting 1 for h and 155 for fh yields $155 = 651 + b$, or $155 = 65 + b$. Subtracting 65 from both sides of this equation yields $90 = b$. Substituting 90 for b in the equation $fh = 65h + b$ yields $f(h) = 65h + 90$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.11 — C

Choice C is correct. In the xy-plane, an equation of the graph of a linear function can be written in the form $fx = mx + b$, where m represents the slope and $0, b$ represents the y-intercept of the graph of $y = fx$. It's given that the graph of the linear function f, where $y = fx$, in the xy-plane contains the point $0, 3$. Thus, $b = 3$. The slope of the graph of a line containing any two points x_1, y_1 and x_2, y_2 can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $0, 3$ and $7, 31$, it follows that the slope of the graph of the line containing these points is $m = \frac{31 - 3}{7 - 0}$, or $m = 4$. Substituting 4 for m and 3 for b in $fx = mx + b$ yields $fx = 4x + 3$.

Choice A is incorrect. This function represents a graph with a slope of 28 and a y-intercept of $0, 34$.

Choice B is incorrect. This function represents a graph with a slope of 3 and a y-intercept of $0, 38$.

Choice D is incorrect. This function represents a graph with a slope of 7 and a y-intercept of $0, 3$.

▼ 2.12 — ?

The correct answer is 2. The slope-intercept form of an equation for a line is $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. Two ordered pairs, (x_1, y_1) and (x_2, y_2), can be used to compute the slope using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that $f(0) = 18$ and $f(1) = 20$; therefore, the two ordered pairs for this line are $(0, 18)$ and $(1, 20)$. Substituting these values for (x_1, y_1) and (x_2, y_2) gives $\frac{20 - 18}{1 - 0} = \frac{2}{1}$, or 2.

▼ 2.13 — D

Choice D is correct. It's given that the relationship between x and fx is linear. A linear function can be written in the form $fx = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane. Given two points on a line, x_1, y_1 and x_2, y_2, the slope of the line can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that the company's profit is $\$ 220$ when it sells 8 items and the profit is $\$ 320$ when it sells 10 items. Since fx represents the company's profit, in dollars, for selling x items, the graph of $y = fx$ in the xy-plane passes through the points $8, 220$ and $10, 320$. Substituting $8, 220$ and $10, 320$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{320 - 220}{10 - 8}$, which gives $m = \frac{100}{2}$, or $m = 50$. Substituting 50 for m, 8 for x, and 220 for fx in $fx = mx + b$ yields $220 = 508 + b$, or $220 = 400 + b$. Subtracting 400 from each side of this equation yields $-180 = b$. Substituting 50 for m and -180 for b in $fx = mx + b$ yields $fx = 50x - 180$. Therefore, the equation that defines f is $fx = 50x - 180$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.14 — C

Choice C is correct. It's given that the equation $f(x) = 108 - 9x$ represents the number of free muffins that remain to be given away after x hours. For the given equation $f(12) = 0$, when the value of x is 12, the value of $f(x)$ is 0. It follows that after 12 hours there are 0 free muffins remaining to be given away. Therefore, based on this context, it will take 12 hours to give away all the free muffins.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This represents giving away 12 muffins over 108 hours rather than giving away all 108 muffins in 12 hours.

▼ 2.15 — A

Choice A is correct. The line passes through the origin. Therefore, this is a relationship of the form $d = km$, where k is a constant representing the slope of the graph. To find the value of k, choose a point (m,d) on the graph of the line other than the origin and substitute the values of m and d into the equation. For example, if the point $(2,4)$ is chosen, then $4 = k(2)$, and $k = 2$. Therefore, the equation of the line is $d = 2m$.

Choice B is incorrect and may result from calculating the slope of the line as the change in time over the change in distance traveled instead of the change in distance traveled over the change in time. Choices C and D are incorrect because each of these equations represents a line with a d-intercept of 2. However, the graph shows a line with a d-intercept of 0.

▼ 2.16 — 7

The correct answer is 7. An equation of a line in the *xy*-plane can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the *y*-intercept of the line. It's given that line k has a slope of 5 and a *y*-intercept of $(0, -35)$. Therefore, $m = 5$ and $b = -35$. Substituting 5 for m and -35 for b in the equation $y = mx + b$ yields $y = 5x - 35$. The *x*-intercept of a line in the *xy*-plane is the point where the line intersects the *x*-axis, which is a point with a *y*-coordinate of 0. Substituting 0 for y in the equation $y = 5x - 35$ yields $0 = 5x - 35$. Adding 35 to both sides of this equation yields $35 = 5x$. Dividing both sides of this equation by 5 yields $7 = x$. Therefore, the *x*-coordinate of the *x*-intercept of line k is 7.

▼ 2.17 — D

Choice D is correct. It's given that for the linear function g, the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 32. Substituting 1 for x and 32 for $g(x)$ in $g(x) = mx + b$ yields $32 = m(1) + b$, or $32 = m + b$. Subtracting b from both sides of this equation yields $32 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 28. Substituting 2 for x and 28 for $g(x)$ in $g(x) = mx + b$ yields $28 = m(2) + b$, or $28 = 2m + b$. Substituting $32 - b$ for m in this equation yields $28 = 2(32 - b) + b$. Applying the

distributive property to the right-hand side of this equation yields $28 = 64 - 2b + b$, or $28 = 64 - b$. Subtracting 64 from both sides of this equation yields $-36 = -b$. Dividing both sides of this equation by -1 yields $36 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.18 — D

Choice D is correct. Since f is a linear function, it can be defined by an equation of the form $fx = ax + b$, where a and b are constants. It's given that $f0 = 8$. Substituting 0 for x and 8 for fx in the equation $fx = ax + b$ yields $8 = a0 + b$, or $8 = b$. Substituting 8 for b in the equation $fx = ax + b$ yields $fx = ax + 8$. It's given that $f1 = 12$. Substituting 1 for x and 12 for fx in the equation $fx = ax + 8$ yields $12 = a1 + 8$, or $12 = a + 8$. Subtracting 8 from both sides of this equation yields $a = 4$. Substituting 4 for a in the equation $fx = ax + 8$ yields $fx = 4x + 8$. Therefore, an equation that defines f is $fx = 4x + 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.19 — A

Choice A is correct. It is given that the relationship between population and year is linear; therefore, the function that models the population t years after 2000 is of the form $P(t) = mt + b$, where m is the slope and b is the population when $t = 0$. In the year 2000, $t = 0$. Therefore, $b = 862$. The slope is given by $m = \frac{P(10) - P(0)}{10 - 0} = \frac{846 - 862}{10 - 0} = \frac{-16}{10} = -1.6$. Therefore, $P(t) = -1.6t + 862$, which is equivalent to the equation in choice A.

Choice B is incorrect and may be the result of incorrectly calculating the slope as just the change in the value of P. Choice C is incorrect and may be the result of the same error as in choice B, in addition to incorrectly using t to represent the year, instead of the number of

years after 2000. Choice D is incorrect and may be the result of incorrectly using t to represent the year instead of the number of years after 2000.

▼ 2.20 — C

Choice C is correct. The value of $f(2)$ can be found by substituting 2 for x in the given equation $fx = x + 7$, which yields $f2 = 2 + 7$, or $f2 = 9$. The value of $g2$ can be found by substituting 2 for x in the given equation $gx = 7x$, which yields $g2 = 72$, or $g2 = 14$. The value of the expression $4f2 - g2$ can be found by substituting the corresponding values into the expression, which gives $4(9) - 14$. This expression is equivalent to $36 - 14$, or 22.

Choice A is incorrect. This is the value of $f2 - g2$, not $4f2 - g2$.

Choice B is incorrect and may result from calculating $4f2$ as $4(2) + 7$, rather than $4(2 + 7)$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.21 — D

Choice D is correct. It's given that the function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. The given function $wt = 300 - 4t$ can be rewritten as $wt = -4t + 300$. Thus, for each increase of t by 1, the value of wt decreases by $4(1)$, or 4. Therefore, the predicted volume, in milliliters, draining from the container each second is 4 milliliters.

Choice A is incorrect. This is the amount of liquid, in milliliters, in the container before the liquid begins draining.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 2.22 — A

Choice A is correct. Since h is a linear function, it can be defined by an equation of the form $hx = mx + b$, where m and b are constants. It's given that $h0 = 0$. Substituting 0 for x and 0 for hx in the equation $hx = mx + b$ yields $0 = 0m + b$, or $0 = b$. Substituting 0 for b in the equation $hx = mx + b$ yields $hx = mx$. It's also given that $h11 = 10$. Substituting 11 for x and

10 for hx in the equation $hx = mx$ yields $10 = 11m$. Dividing both sides of this equation by 11 yields $m = \frac{10}{11}$. Substituting $\frac{10}{11}$ for m in the equation $hx = mx$ yields $hx = \frac{10}{11}x$. Therefore, an equation that defines h is $hx = \frac{10}{11}x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.23 — D

Choice D is correct. It's given that $f(x) = x + 6$. Substituting 2 for x in the function f yields $f(2) = 2 + 6$, or $f(2) = 8$. It's also given that $g(x) = 6x$. Substituting 2 for x in the function g yields $g(2) = (6)(2)$, or $g2 = 12$. Substituting 8 for $f(2)$ and 12 for $g(2)$ in the expression $7f(2) - g(2)$ yields $7(8) - 12$, or 44. Therefore, the value of $7f(2) - g(2)$ is 44.

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $7f(2) - g(2)$.

Choice B is incorrect. This is the value of $f(2)$, not $7f(2) - g(2)$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.24 — C

Choice C is correct. It's given that for each point (x, y) on the graph of the given equation, the x-coordinate represents the number of seconds after Sheila applied the brakes, and the y-coordinate represents the speed of Sheila's bicycle at that moment in time. For the graph of the equation, the y-coordinate of the x-intercept is 0. Therefore, the x-coordinate of the x-intercept of the graph of the given equation represents the number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop.

Choice A is incorrect. The speed of Sheila's bicycle before she applied the brakes is represented by the y-coordinate of the y-intercept of the graph of the given equation, not the x-coordinate of the x-intercept. Choice B is incorrect. The number of feet per second the speed of Sheila's bicycle decreased each second after Sheila applied the brakes is represented by the slope of the graph of the given equation, not the x-coordinate of the x-intercept. Choice D is incorrect and may result from misinterpreting x as the distance, in feet, traveled after applying the brakes, rather than the time, in seconds, after applying the brakes.

▼ 2.25 — B

Choice B is correct. The value of $f(9)-f(1)$ can be calculated by finding the values of $f(9)$ and $f(1)$. The value of $f(9)$ can be found by substituting 9 for x in the given function: $f(9)=\frac{(9+7)}{4}$. This equation can be rewritten as $f(9)=\frac{16}{4}$, or 4. Then, the value of $f(1)$ can be found by substituting 1 for x in the given function: $f(1)=\frac{(1+7)}{4}$. This equation can be rewritten as $f(1)=\frac{8}{4}$, or 2. Therefore, $f(9)-f(1)=4-2$, which is equivalent to 2.

Choices A, C, and D are incorrect and may result from incorrectly substituting values of x in the given function or making computational errors.

▼ 2.26 — A

Choice A is correct. The *x*-intercept of a graph in the *xy*-plane is the point on the graph where $y = 0$. It's given that function *h* is defined by $hx = 4x + 28$. Therefore, the equation representing the graph of $y = hx$ is $y = 4x + 28$. Substituting 0 for *y* in the equation $y = 4x + 28$ yields $0 = 4x + 28$. Subtracting 28 from both sides of this equation yields $-28 = 4x$. Dividing both sides of this equation by 4 yields $-7 = x$. Therefore, the *x*-intercept of the graph of $y = hx$ in the *xy*-plane is $-7, 0$. It's given that the *x*-intercept of the graph of $y = hx$ is $a, 0$. Therefore, $a = -7$. The *y*-intercept of a graph in the *xy*-plane is the point on the graph where $x = 0$. Substituting 0 for *x* in the equation $y = 4x + 28$ yields $y = 40 + 28$, or $y = 28$. Therefore, the *y*-intercept of the graph of $y = hx$ in the *xy*-plane is $0, 28$. It's given that the *y*-intercept of the graph of $y = hx$ is $0, b$. Therefore, $b = 28$. If $a = -7$ and $b = 28$, then the value of $a + b$ is $-7 + 28$, or 21.

Choice B is incorrect. This is the value of *b*, not $a + b$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $-a + b$, not $a + b$.

▼ 2.27 — A

Choice A is correct. The x-intercept of the line with equation y = 120 – 25x can be found by substituting 0 for y and finding the value of x. When y = 0, x = 4.8, so the x-intercept is at (4.8, 0). Since y represents the number of tons of cargo remaining to be moved x hours after the team started working, it follows that the x-intercept refers to the team having no cargo remaining to be moved after 4.8 hours. In other words, the team will have moved all of the cargo after about 4.8 hours.

Choice B is incorrect and may result from incorrectly interpreting the value 4.8. Choices C and D are incorrect and may result from misunderstanding the x-intercept. These statements are accurate but not directly relevant to the x-intercept.

▼ 2.28 — B

Choice B is correct. The relationship between x food calories and k kilojoules can be modeled as a proportional relationship. Let (x_1,k_1) and (x_2,k_2) represent the values in the first two rows in the table: $(4.0,16.7)$ and $(9.0,37.7)$. The rate of change, or $\frac{(k_2-k_1)}{(x_2-x_1)}$, is $\frac{21}{5}=4.2$; therefore, the equation that best represents the relationship between x and k is $k=4.2x$.

Choice A is incorrect and may be the result of calculating the rate of change using $\frac{(x_2-x_1)}{(k_2-k_1)}$.

Choice C is incorrect because the number of kilojoules is greater than the number of food calories. Choice D is incorrect and may be the result of an error when setting up the equation.

▼ 2.29 — C

Choice C is correct. An equation defining a linear function can be written in the form $gx = mx + b$, where m is the slope and $0, b$ is the y-intercept of the graph of $y = gx$ in the xy-plane. It's given that the graph of $y = gx$ has a slope of 2. Therefore, $m = 2$. It's also given that the graph of $y = gx$ passes through the point $1, 14$. It follows that when $x = 1$, $gx = 14$. Substituting 1 for x, 14 for gx, and 2 for m in the equation $gx = mx + b$ yields $14 = 21 + b$, or $14 = 2 + b$. Subtracting 2 from each side of this equation yields $12 = b$. Therefore, $b = 12$.

Substituting 2 for m and 12 for b in the equation $gx = mx + b$ yields $gx = 2x + 12$. Therefore, the equation that defines g is $gx = 2x + 12$.

Choice A is incorrect. For this function, the graph of $y = gx$ in the xy-plane passes through the point $1, 2$, not $1, 14$.

Choice B is incorrect. For this function, the graph of $y = gx$ in the xy-plane passes through the point $1, 4$, not $1, 14$.

Choice D is incorrect. For this function, the graph of $y = gx$ in the xy-plane passes through the point $1, 16$, not $1, 14$.

▼ 2.30 — A

Choice A is correct. For the linear function f, it's given that $f7 = 28$. Substituting 7 for x and 28 for fx in the given function yields $28 = 47 + b$, or $28 = 28 + b$. Subtracting 28 from each side of this equation yields $0 = b$. Therefore, the value of b is 0.

Choice B is incorrect. Substituting 1 for b in the given function yields $fx = 4x + 1$. For this function, when the value of x is 7, the value of fx is 29, not 28.

Choice C is incorrect. Substituting 4 for b in the given function yields $fx = 4x + 4$. For this function, when the value of x is 7, the value of fx is 32, not 28.

Choice D is incorrect. Substituting 7 for b in the given function yields $fx = 4x + 7$. For this function, when the value of x is 7, the value of fx is 35, not 28.

▼ 2.31 — C

Choice C is correct. It's given that sx is the number of spoons remaining in a dispenser x minutes after the dispenser had been loaded with spoons. The expression $s3$ represents the value of sx when $x = 3$. It follows that $s3$ represents the number of spoons remaining in the dispenser 3 minutes after it had been loaded with spoons. Therefore, the best interpretation of the equation $s3 = 121$ is that there were 121 spoons remaining in the dispenser 3 minutes after it had been loaded with spoons.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This would be the best interpretation of $s121 = 3$, not $s3 = 121$.

Choice D is incorrect and may result from conceptual errors.

▼ 2.32 — A

Choice A is correct. It's given that there is a linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. It follows that this relationship can be represented by an equation of the form $p = mc + b$, where m is the rate of change of p in this relationship and b is a constant. The rate of change of p in this relationship can be calculated by dividing the difference in any two values of p by the difference in the corresponding values of c. Using two pairs of values given in the table, the rate of change of p in this relationship is $\frac{284 - 174}{5 - 3}$, or 55. Substituting 55 for m in the equation $p = mc + b$ yields $p = 55c + b$. The value of b can be found by substituting any value of c and its corresponding value of p for c and p, respectively, in this equation. Substituting 10 for c and 559 for p yields $559 = 5510 + b$, or $559 = 550 + b$. Subtracting 550 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $p = 55c + b$ yields $p = 55c + 9$. Subtracting 9 from both sides of this equation yields $p - 9 = 55c$. Subtracting p from both sides of this equation yields $-9 = 55c - p$, or $55c - p = -9$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.33 — C

Choice C is correct. It's given that the relationship between x and y is linear. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of the relationship in the xy-plane. It's given that for every increase in the value of x by 1, the value of y increases by 8. The slope of a line can be expressed as the change in y over the change in x. Thus, the slope, m, of the line representing this relationship can be expressed as $\frac{8}{1}$, or 8. Substituting 8 for m in the equation $y = mx + b$ yields $y = 8x + b$. It's also given that when the value of x is 2, the value of y is 18. Substituting 2 for x and 18 for y in the equation $y = 8x + b$ yields $18 = 82 + b$, or $18 = 16 + b$. Subtracting 16 from each side of this equation yields $2 = b$. Substituting 2 for b in the equation $y = 8x + b$ yields $y = 8x + 2$. Therefore, the equation $y = 8x + 2$ represents this relationship.

Choice A is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 22, not 18.

Choice B is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 12, not 18.

Choice D is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 3, not 8, and when the value of x is 2, the value of y is 32, not 18.

▼ 2.34 — 9

The correct answer is 9. The y-intercept of the graph of $y = f(x)$ in the xy-plane is the point where the graph of $y = f(x)$ crosses the y-axis, which occurs at $x = 0$. It's given that the function f is defined by $f(x) = -9x + 9$. Substituting 0 for x and y for $f(x)$ in this equation yields $y = -9(0) + 9$, or $y = 9$. It follows that $y = 9$ when $x = 0$ and that the y-intercept of the graph of $y = f(x)$ in the xy-plane is $(0, 9)$. Therefore, the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane is 9.

▼ 2.35 — C

Choice C is correct. In the xy-plane, the graph of a linear function can be written in the form $fx = mx + b$, where m represents the slope and $0, b$ represents the y-intercept of the graph of $y = fx$. It's given that the graph of the linear function f, where $y = fx$, in the xy-plane contains the point $0, 2$. Thus, $b = 2$. The slope of the graph of a line containing any two points x_1, y_1 and x_2, y_2 can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $0, 2$ and $8, 34$, it follows that the slope of the graph of the line containing these points is $m = \frac{34 - 2}{8 - 0}$, or $m = 4$. Substituting 4 for m and 2 for b in $fx = mx + b$ yields $fx = 4x + 2$.

Choice A is incorrect. This function represents a graph with a slope of 2 and a y-intercept of $0, 42$.

Choice B is incorrect. This function represents a graph with a slope of 32 and a y-intercept of $0, 36$.

Choice D is incorrect. This function represents a graph with a slope of 8 and a y-intercept of $0, 2$.

▼ 2.36 — B

Choice B is correct. For the given linear function f, fx must equal 39 for all values of x. Of the given choices, only choice B gives three values of x and their corresponding values of fx for the given linear function f.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.37 — C

Choice C is correct. The average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers, is given in the table. This linear relationship can be represented by an equation in the form $t = ad + b$, where a and b are constants. The table shows that it takes Carly an average time of 8 minutes to travel 0.32 kilometers. Substituting 8 for t and 0.32 for d in the equation $t = ad + b$ yields $8 = 0.32a + b$. Subtracting $0.32a$ from both sides of this equation yields $8 - 0.32a = b$. The table also shows that it takes Carly an average time of 14 minutes to travel 0.56 kilometers. Substituting 14 for t and 0.56 for d in the equation $t = ad + b$ yields $14 = 0.56a + b$. Subtracting $0.56a$ from both sides of this equation yields $14 - 0.56a = b$. Substituting $8 - 0.32a$ for b in this equation yields $14 - 0.56a = 8 - 0.32a$. Subtracting 8 from both sides of this equation yields $6 - 0.56a = -0.32a$. Adding $0.56a$ to both sides of this equation yields $6 = 0.24a$. Dividing both sides of this equation by 0.24 yields $25 = a$. Substituting 25 for a in the equation $8 = 0.32a + b$ yields $8 = 0.32(25) + b$, or $8 = 8 + b$. Subtracting 8 from both sides of this equation yields $0 = b$. Substituting 25 for a and 0 for b in the equation $t = ad + b$ yields $t = 25d + 0$, or $t = 25d$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.38 — D

Choice D is correct. It's given that for the linear function g, the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 54. Substituting 1 for x and 54 for $g(x)$ in $g(x) = mx + b$ yields $54 = m(1) + b$ or, $54 = m + b$. Subtracting b from both sides of this equation yields

$54 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 51. Substituting 2 for x and 51 for $g(x)$ in $g(x) = mx + b$ yields $51 = m(2) + b$, or $51 = 2m + b$. Substituting $54 - b$ for m in this equation yields $51 = 2(54 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $51 = 108 - 2b + b$, or $51 = 108 - b$. Subtracting 108 from both sides of this equation yields $-57 = -b$. Dividing both sides of this equation by -1 yields $57 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.39 — 241

The correct answer is 241. For a certain animal, it's given that a model predicts the animal weighed 241 pounds when it was born and gained 3 pounds per day in its first year of life. It's also given that this model is defined by an equation in the form $fx = a + bx$, where fx is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. It follows that a represents the predicted weight, in pounds, of the animal when it was born and b represents the predicted rate of weight gain, in pounds per day, in its first year of life. Thus, the value of a is 241.

▼ 2.40 — 39

The correct answer is 39. It's given that for the linear function j, m is a constant and $j12 = 18$. Substituting 12 for x and 18 for jx in the given equation yields $18 = m12 + 144$. Subtracting 144 from both sides of this equation yields $-126 = m12$. Dividing both sides of this equation by 12 yields $-10.5 = m$. Substituting -10.5 for m in the given equation, $jx = mx + 144$, yields $jx = -10.5x + 144$. Substituting 10 for x in this equation yields $j10 = -10.510 + 144$, or $j10 = 39$. Therefore, the value of $j10$ is 39.

▼ 2.41 — B

Choice B is correct. It's given that $fx = 2x + 244$ represents the perimeter, in centimeters (cm), of a rectangle with a length of x cm and a fixed width. If w represents a fixed width, in cm, then the perimeter, in cm, of a rectangle with a length of x cm and a fixed width of w cm can be given by the function $fx = 2x + 2w$. Therefore, $2x + 2w = 2x + 244$. Subtracting $2x$ from both sides of this equation yields $2w = 244$. Dividing both sides of this equation by 2 yields $w = 122$. Therefore, the width, in cm, of the rectangle is 122.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.42 — C

Choice C is correct. A linear function can be written in the form $f(x)=mx+b$, where m is the slope and b is the y-coordinate of the y-intercept of the line. The slope of the graph can be found using the formula $m=\frac{y_2-y_1}{x_2-x_1}$. Substituting the values of the given points into this formula yields $m=\frac{-5-3}{4-(-2)}$ or $m=\frac{-8}{6}$, which simplifies to $m=-\frac{4}{3}$. Only choice C shows an equation with this slope.

Choices A, B, and D are incorrect and may result from computation errors or misinterpreting the given information.

▼ 2.43 — 2850

The correct answer is 2,850. It's given that the function $fx = 45x + 600$ gives the monthly fee, in dollars, a facility charges to keep x crates in storage. Substituting 50 for x in this function yields $f50 = 4550 + 600$, or $f50 = 2{,}850$. Therefore, the monthly fee, in dollars, the facility charges to keep 50 crates in storage is 2,850.

▼ 2.44 — B

Choice B is correct. A linear function has a constant rate of change, and any two rows of the table shown can be used to calculate this rate. From the first row to the second, the value of x is increased by 2 and the value of $f(x)$ is increased by $6=4-(-2)$. So the values of $f(x)$ increase by 3 for every increase by 1 in the value of x. Since $f(2)=4$, it follows that $f(2+1)=4+3=7$. Therefore, $f(3)=7$.

Choice A is incorrect. This is the third x-value in the table, not $f(3)$. Choices C and D are incorrect and may result from errors when calculating the function's rate of change.

▼ 2.45 — -42

The correct answer is -42. For a linear equation in the form $y = mx + b$, m represents the slope and b represents the y-coordinate of the y-intercept of the graph in the xy-plane. The slope of the graph of a line containing any two points, (x_1, y_1) and (x_2, y_2), can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that in the xy-plane, line k passes through the points $(8, 6)$ and $(9, 12)$. Substituting $(8, 6)$ and $(9, 12)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{12 - 6}{9 - 8}$, or $m = 6$. Since line k passes through the point $(9, 12)$ and the slope of the line is 6, substituting 9 for x, 12 for y, and 6 for m in the equation $y = mx + b$ yields $12 = (6)(9) + b$, or $12 = 54 + b$. Subtracting 54 from both sides of this equation yields $-42 = b$. Therefore, the value of b is -42.

▼ 2.46 — D

Choice D is correct. It's given that the boiling point of water at sea level is 212°F and that for every 550 feet above sea level, the boiling point of water is lowered by about 1°F. Therefore, the change in the boiling point of water x feet above sea level is represented by the expression $-\frac{x}{550}$. Adding this expression to the boiling point of water at sea level gives the equation for the boiling point B of water, in °F, x feet above sea level: $B = -\frac{x}{550} + 212$, or $B = 212 - \frac{x}{550}$.

Choices A and B are incorrect and may result from using the boiling point of water at sea level as the rate of change and the rate of change as the initial boiling point of water at sea level. Choice C is incorrect and may result from representing the change in the boiling point of water as an increase rather than a decrease.

▼ 2.47 — 67

The correct answer is 67. It's given that $f5 = 32$. Therefore, for the given function f, when $x = 5$, $fx = 32$. Substituting 5 for x and 32 for fx in the given function $fx = 4x + kx - 1$ yields $32 = 45 + k5 - 1$, or $32 = 20 + 4k$. Subtracting 20 from each side of this equation yields $12 = 4k$. Dividing each side of this equation by 4 yields $k = 3$. Substituting 3 for k in the given function $fx = 4x + kx - 1$ yields $fx = 4x + 3x - 1$, which is equivalent to $fx = 4x + 3x - 3$, or $fx = 7x - 3$. Substituting 10 for x into this equation yields $f10 = 710 - 3$, or $f10 = 67$. Therefore, the value of $f10$ is 67.

▼ 2.48 — ?

The correct answer is 2.6. According to the model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to 4 times the body weight, in grams, of the bee. Let x represent the body weight, in grams, of a worker bumblebee and let y represent the head width, in millimeters. Translating the verbal description of the model into an equation yields $y = 0.6 + 4x$. Substituting 0.5 grams for x in this equation yields $y = 0.6 + 4(0.5)$, or $y = 2.6$. Therefore, a worker bumblebee with a body weight of 0.5 grams has an estimated head width of 2.6 millimeters. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

▼ 2.49 — D

Choice D is correct. The given function f is a linear function. Therefore, the graph of $y = fx$ in the xy-plane has one x-intercept at the point $k, 0$, where k is a constant. Substituting 0 for fx and k for x in the given function yields $0 = 7k - 84$. Adding 84 to both sides of this equation yields $84 = 7k$. Dividing both sides of this equation by 7 yields $12 = k$. Therefore, the x-intercept of the graph of $y = fx$ in the xy-plane is $12, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.50 — 774

The correct answer is 774. It's given that Brian saves $\frac{2}{5}$ of the $\$ 215$ he earns each week from his job. Therefore, Brian saves $\frac{2}{5} \$ 215$, or $\$ 86$, per week. If Brian continues to save at this rate of $\$ 86$ per week for 9 weeks, then he will save a total of 986, or 774, dollars.

▼ 2.51 — 11

The correct answer is 11. It's given that fx is defined by the equation $f(x) = mx - 28$, where m is a constant. It's also given in the table that when $x = 10$, $fx = 82$. Substituting 10 for x and 82 for fx in the equation $fx = mx - 28$ yields, $82 = m10 - 28$. Adding 28 to both sides of this equation yields $110 = 10m$. Dividing both sides of this equation by 10 yields $11 = m$. Therefore, the value of m is 11.

▼ 2.52 — A

Choice A is correct. It's given that the electrician charges a onetime fee plus an hourly rate. It's also given that the graph represents the total charge, in dollars, for x hours of work. This graph shows a linear relationship in the xy-plane. Thus, the total charge y, in dollars, for x hours of work can be represented as $y = mx + b$, where m is the slope and $0, b$ is the y-intercept of the graph of the equation in the xy-plane. Since the given graph represents the total charge, in dollars, by an electrician for x hours of work, it follows that its slope is m, or the electrician's hourly rate.

Choice B is incorrect. The electrician's onetime fee is represented by the y-coordinate of the y-intercept, not the slope, of the graph.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.53 — A

Choice A is correct. It's given that $g(x) = 3(14x - 15)$. Substituting $314x - 15$ for $g(x)$ in the equation $y = g(x) - 2$ yields $y = 3(14x - 15) - 2$. The y-intercept of the graph of a function in the xy-plane is the point where the function crosses the y-axis, which occurs when $x = 0$. Substituting 0 for x in the equation $y = 3(14x - 15) - 2$ yields $y = 3(14(0) - 15) - 2$. It follows that $y = -45 - 2$, or $y = -47$. Therefore, the y-intercept of the graph of $y = g(x) - 2$ in the xy-plane is $(0, -47)$, and the y-coordinate of the y-intercept of the graph is -47.

Choice B is incorrect. This is the y-coordinate of the y-intercept of the graph of $y = g(x)$ in the xy-plane, not $y = g(x) - 2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.54 — .0465, 2/43

The correct answer is $\frac{2}{43}$. It's given that $f(cx) = x - 8$ for all values of x, where c is a positive constant, and $f(2) = 35$. Therefore, for the given function f, $cx = 2$. Dividing both sides of this equation by c yields $x = \frac{2}{c}$. Substituting $\frac{2}{c}$ for x in the equation $f(cx) = x - 8$ yields $f(\frac{2c}{c}) = \frac{2}{c} - 8$, or $f(2) = \frac{2}{c} - 8$. Since it's given that $f(2) = 35$, substituting 35 for $f(2)$ yields $35 = \frac{2}{c} - 8$. Adding 8 to both sides of this equation yields $43 = \frac{2}{c}$. Multiplying both sides of this equation by c yields $43c = 2$. Dividing both sides of this equation by 43 yields $c = \frac{2}{43}$. Note that 2/43, .0465, 0.046, and 0.047 are examples of ways to enter a correct answer.

▼ 2.55 — 515

The correct answer is 515. It's given that the equation $h = \frac{9(v - 273.15)}{5} + 32$ gives the corresponding temperature h, in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. Substituting 467.33 for h in the given equation yields $467.33 = \frac{9(v - 273.15)}{5} + 32$. Subtracting 32 from both sides of this equation yields $435.33 = \frac{9(v - 273.15)}{5}$. Multiplying both sides of this equation by 5 yields $2{,}176.65 = 9(v - 273.15)$. Dividing both sides of this equation by 9 yields $241.85 = v - 273.15$. Adding 273.15 to both sides of this equation yields $515 = v$. Therefore, if a substance has a temperature of 467.33 degrees Fahrenheit, the corresponding temperature, in kelvins, of this substance is 515.

▼ 2.56 — B

Choice B is correct. It's given that the regular price of a session at a yoga studio is $\$ 25.40$. It's also given that the yoga studio is offering a deal where the first session is free, or $\$ 0$; that the second session is half off the regular price, or $\$ 25.40 - \frac{1}{2}(\$ 25.40)$, which is equal to $\$ 12.70$; and that the remaining sessions are at the regular price, $\$ 25.40$. Since x represents the total number of sessions and two sessions have already been accounted for, the number of remaining sessions can be represented by $x - 2$. The total cost, in dollars, of purchasing x sessions using this deal can be represented by the function $f(x) = 0 + 12.70 + 25.40(x - 2)$, or $f(x) = 25.40(x - 2) + 12.70$.

Choice A is incorrect. The function represents the total cost of purchasing x sessions when the first session is half off the regular price and the remaining sessions are at the regular price.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.57 — A

Choice A is correct. Let the economist's model be the linear function $Q = mP + b$, where Q is the demand, P is the selling price, m is the slope of the line, and b is the y-coordinate of the y-intercept of the line in the xy-plane, where $y = Q$. Two pairs of the selling price P and the demand Q are given. Using the coordinate pairs (P, Q), two points that satisfy the function are $(40, 20{,}000)$ and $(60, 15{,}000)$. The slope m of the function can be found using the formula $m = \frac{Q_2 - Q_1}{P_2 - P_1}$. Substituting the given values into this formula yields

$m = \frac{15{,}000 - 20{,}000}{60 - 40}$, or $m = -250$. Therefore, $Q = -250P + b$. The value of b can be found by substituting one of the points into the function. Substituting the values of P and Q from the point $(40{,}20{,}000)$ yields $20{,}000 = -250(40) + b$, or $20{,}000 = -10{,}000 + b$. Adding 10,000 to both sides of this equation yields $b = 30{,}000$. Therefore, the linear function the economist used as the model is $Q = -250P + 30{,}000$. Substituting 55 for P yields $Q = -250(55) + 30{,}000 = 16{,}250$. It follows that when the selling price is $55 per unit, the demand is 16,250 units.

Choices B, C, and D are incorrect and may result from calculation or conceptual errors.

▼ 2.58 — A

Choice A is correct. It's given that the window repair specialist charges $\$ 220$ for the first two hours of repair plus an hourly fee for each additional hour. Let n represent the hourly fee for each additional hour after the first two hours. Since it's given that x is the number of hours of repair, it follows that the charge generated by the hourly fee after the first two hours can be represented by the expression $nx - 2$. Therefore, the total cost, in dollars, for x hours of repair is $fx = 220 + nx - 2$. It's given that the total cost for 5 hours of repair is $\$ 400$. Substituting 5 for x and 400 for fx into the equation $fx = 220 + nx - 2$ yields $400 = 220 + n5 - 2$, or $400 = 220 + 3n$. Subtracting 220 from both sides of this equation yields $180 = 3n$. Dividing both sides of this equation by 3 yields $n = 60$. Substituting 60 for n in the equation $fx = 220 + nx - 2$ yields $fx = 220 + 60x - 2$, which is equivalent to $fx = 220 + 60x - 120$, or $fx = 60x + 100$. Therefore, the total cost, in dollars, for x hours of repair is $fx = 60x + 100$.

Choice B is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges $\$ 340$, rather than $\$ 220$, for the first two hours of repair.

Choice C is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges $\$ 160$, rather than $\$ 220$, for the first two hours of repair, and an hourly fee of $\$ 80$, rather than $\$ 60$, after the first two hours.

Choice D is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges $\$ 380$, rather than $\$ 220$, for the first two hours of repair, and an hourly fee of $\$ 80$, rather than $\$ 60$, after the first two hours.

▼ 2.59 — B

Choice B is correct. It's given that $f(3x) = x - 6$ for all values of x. If $3x = 6$, then $f(3x)$ will equal $f(6)$. Dividing both sides of $3x = 6$ by 3 gives $x = 2$. Therefore, substituting 2 for x in the given equation yields $f(3 \times 2) = 2 - 6$, which can be rewritten as $f(6) = -4$.

Choice A is incorrect. This is the value of the constant in the given equation for f. Choice C is incorrect and may result from substituting $x = 6$, rather than $x = 2$, into the given equation. Choice D is incorrect. This is the value of x that yields $f(6)$ for the left-hand side of the given equation; it's not the value of $f(6)$.

▼ 2.60 — D

Choice D is correct. The table gives that $f(x) = 0$ when $x = 2$. Substituting 0 for fx and 2 for x into the equation $f(x) = ax + b$ yields $0 = 2a + b$. Subtracting $2a$ from both sides of this equation yields $b = -2a$. The table gives that $f(x) = -64$ when $x = 1$. Substituting $-2a$ for b, -64 for fx, and 1 for x into the equation $f(x) = ax + b$ yields $-64 = a1 + -2a$. Combining like terms yields $-64 = -a$, or $a = 64$. Since $b = -2a$, substituting 64 for a into this equation gives $b = -264$, which yields $b = -128$. Thus, the value of $a - b$ can be written as $64 - (-128)$, which is 192.

Choice A is incorrect. This is the value of $a + b$, not $a - b$.

Choice B is incorrect. This is the value of $a - 2$, not $a - b$.

Choice C is incorrect. This is the value of $2a$, not $a - b$.

▼ 2.61 — 3331

The correct answer is 3,331. For engine speeds between 1,000 revolutions per minute (rpm) and 6,000 rpm, it's given that the equation $f(x) = \frac{1}{7}(x - a) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (bhp), at an engine speed of x rpm, where a is a constant. It's also given that the car's predicted power is 433 bhp at an engine speed of 3,331 rpm. Substituting 3,331 for x and 433 for $f(x)$ in the equation yields $433 = \frac{1}{7}(3{,}331 - a) + 433$. Subtracting 433 from both sides of this equation yields $0 = \frac{1}{7}(3{,}331 - a)$. Multiplying both sides of this equation by 7 yields $0 = 3{,}331 - a$. Adding a to both sides of this equation yields $a = 3{,}331$. Thus, the value of a is 3,331.

▼ 2.62 — A

Choice A is correct. It's given that the cost of renting a carpet cleaner is $\$52$ for the first day and $\$26$ for each additional day. Therefore, the cost Cd, in dollars, of renting the carpet cleaner for d days is the sum of the cost for the first day, $\$52$, and the cost for the additional $d - 1$ days, $\$26(d - 1)$. It follows that $Cd = 52 + 26(d - 1)$, which is equivalent to $Cd = 52 + 26d - 26$, or $Cd = 26d + 26$.

Choice B is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$78$, not $\$52$, for the first day and $\$26$ for each additional day.

Choice C is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$26$, not $\$52$, for the first day and $\$52$, not $\$26$, for each additional day.

Choice D is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$130$, not $\$52$, for the first day and $\$52$, not $\$26$, for each additional day.

▼ 2.63 — B

Choice B is correct. It's given that a train traveled 173 miles in the first 3 hours of a trip. It's also given that the train traveled at an average speed of 53 miles per hour for the remainder of the trip. Since the trip was x hours long, it follows that the remainder of the trip was $x - 3$ hours. Therefore, $53(x - 3)$ represents the distance, in miles, the train traveled for the remainder of the trip. Adding 173 to $53(x - 3)$ gives the total number of miles, y, the train traveled if the trip was x hours long and $x > 3$. It follows that $y = 173 + 53(x - 3)$, which is equivalent to $y = 173 + 53x - 159$, or $y = 53x + 14$.

Choice A is incorrect. This equation gives the total number of miles, y, the train traveled if the trip was $x + 3$, not x, hours long.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.64 — B

Choice B is correct. The graph of a line in the xy-plane can be represented by the equation $y = mx + b$, where m is the slope of the line and $0, b$ is the y-intercept. The slope of a line that passes through the points x_1, y_1 and x_2, y_2 can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $-1, -2$ and $0, -4$. Substituting $-1, -2$ and $0, 4$ for x_1, y_1 and x_2, y_2, respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{-4 - -2}{0 - -1}$, which is

equivalent to $m = \frac{-2}{1}$, or $m = -2$. Since the line shown passes through the point $0, -4$, it follows that $b = -4$. Substituting -2 for m and -4 for b in the equation $y = mx + b$ yields $y = -2x - 4$. It's given that the graph shown is the graph of $y = fx - 11$. Substituting $-2x - 4$ for y in the equation $y = fx - 11$ yields $-2x - 4 = fx - 11$. Adding 11 to both sides of this equation yields $-2x + 7 = fx$. Therefore, the equation $fx = -2x + 7$ defines the linear function f.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.65 — A

Choice A is correct. It's given that the graph of the linear function $y = fx + 19$ is shown. This means that the graph of $y = fx + 19$ can be translated down 19 units to create the graph of $y = fx$ and the y-coordinate of every point on the graph of $y = fx + 19$ can be decreased by 19 to find the resulting point on the graph of $y = fx$. The y-intercept of the graph of $y = fx + 19$ is $0, 3$. Translating the graph of $y = fx + 19$ down 19 units results in a y-intercept of the graph of $y = fx$ at the point $0, 3 - 19$, or $0, -16$. The graph of $y = fx + 19$ slants down from left to right, so the slope of the graph is negative. The translation of a linear graph changes its position, but does not change its slope. It follows that the slope of the graph of $y = fx$ is also negative. The equation of a linear function f can be written in the form $fx = b + mx$, where b is the y-coordinate of the y-intercept and m is the slope of the graph of $y = fx$. It's given that c and d are positive constants. Since the y-coordinate of the y-intercept and the slope of the graph of $y = fx$ are both negative, it follows that $fx = -d - cx$ could define f.

Choice B is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept.

Choice C is incorrect. This could define a linear function where its graph has a positive, not negative, slope.

Choice D is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept and a positive, not negative, slope.

▼ 2.66 — 500

The correct answer is 500. It's given that the graph shows the total distance y, in meters, Mary had walked on the trail x minutes after leaving the trail sign. The graph represents a

linear relationship between time and distance, so the distance walked in a given time interval corresponds to the change in the *y*-values over that interval. At $x = 0$, the graph shows a *y*-value of 150, and at $x = 10$, the graph shows a *y*-value of 650. The distance Mary walked in the 10 minutes after leaving the trail sign is the difference between these *y*-values, which can be written as $650 - 150$, or 500. Therefore, the distance, in meters, Mary walked in the 10 minutes after leaving the trail sign is 500.

▼ 2.67 — 257

The correct answer is 257. It's given that a linear model estimates the population of a city from 1991 to 2015. Since the population can be estimated using a linear model, it follows that there is a constant rate of change for the model. It's also given that the model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and x thousand in 2015. The change in the population between 2011 and 1991 is $224 - 57$, or 167, thousand. The change in the number of years between 2011 and 1991 is $2011 - 1991$, or 20, years. Dividing 167 by 20 gives $167 / 20$, or 8.35, thousand per year. Thus, the change in population per year from 1991 to 2015 estimated by the model is 8.35 thousand. The change in the number of years between 2015 and 2011 is $2015 - 2011$, or 4, years. Multiplying the change in population per year by the change in number of years yields the increase in population from 2011 to 2015 estimated by the model: 8.354, or 33.4, thousand. Adding the change in population from 2011 to 2015 estimated by the model to the estimated population in 2011 yields the estimated population in 2015. Thus, the estimated population in 2015 is $33.4 + 224$, or 257.4, thousand. Therefore to the nearest whole number, the value of x is 257.

▼ 2.68 — A

Choice A is correct. It's given that the function $Fx = \frac{9}{5}x - 273.15 + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by 2.10 kelvins means that the value of x increased by 2.10 kelvins. It follows that an increase in x by 2.10 increases $F(x)$ by $\frac{9}{5}2.10$, or 3.78. Therefore, if a temperature increased by 2.10 kelvins, the temperature increased by 3.78 degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.69 — C

Choice C is correct. It is assumed that the oil and gas production decreased at a constant rate. Therefore, the function f that best models the production t years after the year 2000 can be written as a linear function, $f(t)=mt+b$, where m is the rate of change of the oil and gas production and b is the oil and gas production, in millions of barrels, in the year 2000. Since there were 4 million barrels of oil and gas produced in 2000, $b=4$. The rate of change, m, can be calculated as $\frac{4-1.9}{0-13}=-\frac{2.1}{13}$, which is equivalent to $-\frac{21}{130}$, the rate of change in choice C.

Choices A and B are incorrect because each of these functions has a positive rate of change. Since the oil and gas production decreased over time, the rate of change must be negative. Choice D is incorrect. This model may result from misinterpreting 1.9 million barrels as the amount by which the production decreased.

▼ 2.70 — B

Choice B is correct. The equation of a linear function can be written in the form $y=mx+b$, where $y=f(x)$, m is the slope of the graph of $y=f(x)$, and b is the y-coordinate of the y-intercept of the graph. The value of m can be found using the slope formula, $m=\frac{y_2-y_1}{x_2-x_1}$. According to the table, the points $(-11,21)$ and $(-10,18)$ lie on the graph of $y=f(x)$. Using these two points in the slope formula yields $m=\frac{18-21}{-10+11}$, or -3. Substituting -3 for m in the slope-intercept form of the equation yields $y=-3x+b$. The value of b can be found by substituting values from the table and solving; for example, substituting the coordinates of the point $(-11,21)$ into the equation $y=-3x+b$ gives $21=-3(-11)+b$, which yields $b=-12$. This means the function given by the table can be represented by the equation $y=-3x-12$. The value of the x-intercept of the graph of $y=f(x)$ can be determined by finding the value of x when $y=0$. Substituting $y=0$ into $y=-3x-12$ yields $0=-3x-12$, or $x=-4$. This corresponds to the point $(-4,0)$.

Choice A is incorrect and may result from substituting the value of the slope for the x-coordinate of the x-intercept. Choice C is incorrect and may result from a calculation error. Choice D is incorrect and may result from substituting the y-coordinate of the y-intercept for the x-coordinate of the x-intercept.

▼ 2.71 — A

Choice A is correct. It's given that the cost of renting a large canopy tent is $\$430$ for the first day and $\$215$ for each additional day for up to 10 days. For x days of renting the tent, the cost includes $\$430$ for the first day and $\$215$ for each of the $(x - 1)$ additional days. It follows that the cost y, in dollars, of renting the tent can be expressed as $y = 430 + 215(x - 1)$, which is equivalent to $y = 430 + 215x - 215$, or $y = 215x + 215$. Therefore, the equation $y = 215x + 215$ gives the cost of renting the tent for x days, where x is a positive integer and $x \leq 10$.

Choice B is incorrect. This equation represents a situation where the cost of renting the tent for the first day is $\$215$, not $\$430$, and the cost for each additional day is $\$430$, not $\$215$.

Choice C is incorrect. This equation represents a situation where the cost of renting the tent for the first day is $\$645$, not $\$430$, and the cost for each additional day is $\$430$, not $\$215$.

Choice D is incorrect. This equation represents a situation where the cost of renting the tent for the first day is $\$645$, not $\$430$.

▼ 2.72 — A

Choice A is correct. It's given that w represents the total wall area, in square feet. Since the walls of the room will be painted twice, the amount of paint, in gallons, needs to cover $2w$ square feet. It's also given that one gallon of paint will cover 220 square feet. Dividing the total area, in square feet, of the surface to be painted by the number of square feet covered by one gallon of paint gives the number of gallons of paint that will be needed. Dividing $2w$ by 220 yields $\frac{2w}{220}$, or $\frac{w}{110}$. Therefore, the equation that represents the total amount of paint P, in gallons, needed to paint the walls of the room twice is $P = \frac{w}{110}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from finding the amount of paint needed to paint the walls once rather than twice.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.73 — C

Choice C is correct. It's given that $gc + 7 = \frac{c}{4}$. Therefore, for the given linear function g, when $x = c + 7$, $gx = \frac{c}{4}$. Substituting $c + 7$ for x and $\frac{c}{4}$ for gx in $gx = b - 15x$ yields $\frac{c}{4} = b - 15c + 7$. Applying the distributive property to the right-hand side of this equation yields $\frac{c}{4} = b - 15c - 105$. Adding $15c$ to both sides of this equation yields $\frac{c}{4} + 15c = b - 105$. Adding 105 to both sides of this equation yields $\frac{c}{4} + 15c + 105 = b$, or $\frac{61c}{4} + 105 = b$. Therefore, the expression that represents the value of b is $\frac{61c}{4} + 105$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.74 — D

Choice D is correct. Since 2.74 is a constant term, it represents an actual price of gas rather than a measure of change in gas price. To determine what gas price it represents, find x such that F(x) = 2.74, or 2.74 = 2.74 – 0.19(x – 3). Subtracting 2.74 from both sides gives 0 = –0.19(x – 3). Dividing both sides by –0.19 results in 0 = x – 3, or x = 3. Therefore, the average price of gas is $2.74 per gallon 3 months after September 1, 2014, which is December 1, 2014.

Choice A is incorrect. Since 2.74 is a constant, not a multiple of x, it cannot represent a rate of change in price. Choice B is incorrect. The difference in the average price from September 1, 2014, to December 1, 2014, is F(3) – F(0) = 2.74 – 0.19(3 – 3) – (2.74 – 0.19(0 – 3)) = 2.74 – (2.74 + 0.57) = –0.57, which is not 2.74. Choice C is incorrect. The average price per gallon on September 1, 2014, is F(0) = 2.74 – 0.19(0 – 3) = 2.74 + 0.57 = 3.31, which is not 2.74.

▼ 2.75 — A

Choice A is correct. It's given that the function $Fx = \frac{9}{5}x - 273.15 + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by 9.10 kelvins means that the value of x increased by 9.10 kelvins. It follows that an increase in x by 9.10 increases $F(x)$ by $\frac{9}{5}9.10$, or 16.38. Therefore, if a temperature increased by 9.10 kelvins, the temperature increased by 16.38 degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.76 — -12

The correct answer is -12. It's given that the functions f and g are defined as $fx = \frac{1}{4}x - 9$ and $gx = \frac{3}{4}x + 21$. If the function h is defined as $hx = fx + gx$, then substituting $\frac{1}{4}x - 9$ for fx and $\frac{3}{4}x + 21$ for gx in this function yields $hx = \frac{1}{4}x - 9 + \frac{3}{4}x + 21$. This can be rewritten as $hx = \frac{4}{4}x + 12$, or $hx = x + 12$. The x-intercept of a graph in the xy-plane is the point on the graph where $y = 0$. The equation representing the graph of $y = hx$ is $y = x + 12$. Substituting 0 for y in this equation yields $0 = x + 12$. Subtracting 12 from both sides of this equation yields $-12 = x$, or $x = -12$. Therefore, the x-coordinate of the x-intercept of the graph of $y = hx$ in the xy-plane is -12.

▼ 2.77 — A

Choice A is correct. A tour group with n people, where $n \geq 25$, can be split into two subgroups: the first 25 people and the additional $n - 25$ people. Since the museum charges $\$21$ per person for the first 25 people and $\$14$ for each additional person, the charge for the first 25 people is $\$21(25)$ and the charge for the additional $n - 25$ people is $\$14(n - 25)$. Therefore, the total charge, in dollars, is given by the function $f(n) = 21(25) + 14(n - 25)$, or $f(n) = 14n + 175$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.78 — D

Choice D is correct. It's given that the cost of renting a backhoe for up to 10 days is $\$270$ for the first day and $\$135$ for each additional day. Therefore, the cost y, in dollars, for x days, where $x \leq 10$, is the sum of the cost for the first day, $\$270$, and the cost for the additional $x - 1$ days, $\$135x - 1$. It follows that $y = 270 + 135x - 1$, which is equivalent to $y = 270 + 135x - 135$, or $y = 135x + 135$.

Choice A is incorrect. This equation represents a situation where the cost of renting a backhoe is $\$135$ for the first day and $\$270$ for each additional day.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.79 — D

Choice D is correct. It's given that w represents the total fence area, in square feet. Since the fence will be stained twice, the amount of stain, in gallons, will need to cover $2w$ square feet. It's also given that one gallon of stain will cover 170 square feet. Dividing the total area, in square feet, of the surface to be stained by the number of square feet covered by one gallon of stain gives the number of gallons of stain that will be needed. Dividing $2w$ by 170 yields $\frac{2w}{170}$, or $\frac{w}{85}$. Therefore, the equation that represents the total amount of stain S, in gallons, needed to stain the fence of the yard twice is $S = \frac{w}{85}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.80 — D

Choice D is correct. It's given that for a linear relationship between x and y, the table gives three values of x and their corresponding values of y. A linear equation can be written in the form $y = mx + b$, where m is the slope and $(0, b)$ is the y-intercept of the graph of the equation in the xy-plane. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(-34, t)$ and $(-17, t + 23)$ for (x_1, y_1) and (x_2, y_2), respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{(t + 23) - t}{(-17) - (-34)}$, which is equivalent to $m = \frac{t + 23 - t}{-17 + 34}$, or $m = \frac{23}{17}$. Based on the table, the graph of this relationship passes through the point $(0, t + 46)$. It follows that $b = t + 46$. Substituting $\frac{23}{17}$ for m and $t + 46$ for b in the equation $y = mx + b$ yields $y = \frac{23}{17}x + t + 46$. Therefore, the equation $y = \frac{23}{17}x + t + 46$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.81 — D

Choice D is correct. The value 2 is multiplied by w, the weight of the object. When the weight is 0, the length is 30 + 2(0) = 30 centimeters. If the weight increases by w newtons, the length increases by 2w centimeters, or 2 centimeters for each one-newton increase in weight.

Choice A is incorrect because this describes the value 30. Choice B is incorrect because 30 represents the length of the spring before it has been stretched. Choice C is incorrect because this describes the value w.

▼ 2.82 — 16.25, 65/4

The correct answer is $\frac{65}{4}$. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m and b are constants. It's given in the table that when $x = -12$, $y = -45$. Substituting -12 for x and -45 for y in the equation $y = mx + b$ yields $-45 = -12m + b$, which can be rewritten as $-45 + 12m = b$. It's also given in the table that when $x = 6$, $y = 45$. Substituting 6 for x and 45 for y in the equation $y = mx + b$ yields $45 = 6m + b$, which can be rewritten as $45 - 6m = b$. Substituting $-45 + 12m$ for b in this equation yields $45 - 6m = -45 + 12m$. Adding $6m$ to both sides of this equation yields $45 = -45 + 18m$. Adding 45 to both sides of this equation yields $90 = 18m$. Dividing both sides of this equation by 18 yields $5 = m$, or $m = 5$. Substituting 5 for m, -12 for x, and -45 for y in the equation $y = mx + b$ yields $-45 = 5(-12) + b$, or $-45 = -60 + b$. Adding 60 to both sides of this equation yields $15 = b$. Therefore, $m = 5$ and $b = 15$. Substituting 5 for m and 15 for b in the equation $y = mx + b$ yields $y = 5x + 15$. Thus, the equation $y = 5x + 15$ represents the linear relationship between x and y. It's also given that the graph of the linear equation representing this relationship passes through the point $\frac{1}{4}, a$. Substituting $\frac{1}{4}$ for x and a for y in the equation $y = 5x + 15$ yields $a = 5\frac{1}{4} + 15$, which is equivalent to $a = \frac{5}{4} + 15$, or $a = \frac{65}{4}$. Note that 65/4 and 16.25 are examples of ways to enter a correct answer.

▼ 2.83 — D

Choice D is correct. It's given that the equation $y - 5x = 6$ represents the relationship between the number of suits that Kaylani made, x, and the total length of fabric she purchased, y, in yards. Adding $5x$ to both sides of the given equation yields $y = 5x + 6$. Since Kaylani made x suits and used 5 yards of fabric to make each suit, the expression $5x$ represents the total amount of fabric she used to make the suits. Since y represents the total length of fabric Kaylani purchased, in yards, it follows from the equation $y = 5x + 6$ that Kaylani purchased $5x$ yards of fabric to make the suits, plus an additional 6 yards of fabric. Therefore, the best interpretation of 6 in this context is that Kaylani purchased 6 yards more fabric than she used to make the suits.

Choice A is incorrect. Kaylani made a total of x suits, not 6 suits.

Choice B is incorrect. Kaylani purchased a total of y yards of fabric, not a total of 6 yards of fabric.

Choice C is incorrect. Kaylani used a total of $5x$ yards of fabric to make the suits, not a total of 6 yards of fabric.

▼ 2.84 — A

Choice A is correct. It's given that the function $f(x)$ is defined as 19 more than 4 times a number x. This can be represented by the equation $f(x) = 4x + 19$. The x-intercept of the graph of $y = f(x)$ in the xy-plane is the point where the graph intersects the x-axis, or the point on the graph where the value of $f(x)$ is equal to 0. Substituting 0 for $f(x)$ in the equation $f(x) = 4x + 19$ yields $0 = 4x + 19$. Subtracting 19 from each side of this equation yields $-19 = 4x$. Dividing each side of this equation by 4 yields $x = -\frac{19}{4}$. Therefore, when $f(x) = 0$, the number is $-\frac{19}{4}$.

Choice B is incorrect. This is the best interpretation of the y-intercept, not the x-intercept, of the graph of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the best interpretation of the slope, not the x-intercept, of the graph of the function.

▼ 2.85 — B

Choice B is correct. An equation that defines a linear function f can be written in the form $fx = mx + b$, where m and b are constants. It's given in the table that when $x = -4$, $fx = 0$. Substituting -4 for x and 0 for fx in the equation $fx = mx + b$ yields $0 = m(-4) + b$, or $0 = -4m + b$. Adding $4m$ to both sides of this equation yields $4m = b$. Substituting $4m$ for b in the equation $fx = mx + b$ yields $fx = mx + 4m$. It's also given in the table that when $x = -\frac{19}{5}$, $fx = 1$. Substituting $-\frac{19}{5}$ for x and 1 for fx in the equation $fx = mx + 4m$ yields $1 = m(-\frac{19}{5}) + 4m$, or $1 = \frac{1}{5}m$. Multiplying both sides of this equation by 5 yields $m = 5$. Substituting 5 for m in the equation $fx = mx + 4m$ yields $fx = 5x + 45$, or $fx = 5x + 20$. If $hx = fx - 13$, substituting $5x + 20$ for fx in this equation yields $hx = 5x + 20 - 13$, or $hx = 5x + 7$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is an equation that defines the linear function f, not h.